

SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

## THIS MONTH IN SCIENCE:

This month, we explore the origins of palaeontology, we look into methods for predicting premature births, we discover the process of ageing, exploring Saturn's moons and more...



## Droplet like it's watt

Researchers have developed an innovative new technique that allows for the harvesting of electricity from falling rainwater. So far, such approaches were not very effective. The new technique leverages periods of continuous rainfall, and can be very beneficial in developing economies. <https://dgit.in/erain>

## WHAT'S NEW

# Resurrected genes show that the last of the Mammoths were sickly

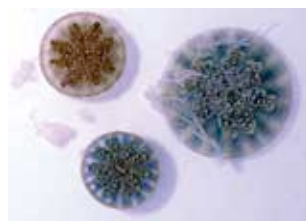
**M**ost woolly mammoths disappeared roughly 10,000 years ago. A small population lived on in Wrangel Island, north of Siberia. Here, mammoths survived until 4,000 years ago. Scientists from the University of Buffalo compared the DNA of these Wrangel Island mammoths with that of mammoth remains elsewhere, as well as living Asian Elephants. They honed in on a few genes that were dramatically different. The mutated genes were then grown in a lab, to check if they behaved normally. The results painted a sorry picture of the last surviving mammoths. They had



lost their sense of smell, their motor functions were severely impaired which means they would have trouble even taking a walk, their neurological development was affected, the bodies could not use insulin effectively, and the males were plagued with infertility. The research shows that the last mammoths suffered from a variety of debilitating genetic conditions, and were an unhealthy population. <https://dgit.in/lastmmth>

## Scientists solve mystery of “stinging water”

**E**xperienced swimmers and divers avoid large populations of Cassiopea jellyfish. They know that the *water* around large groups of these jellyfish can cause stings. They just didn't know why. Theories included severed tentacles of jellyfish, other microscoping stinging organisms, or an outrageous creature called “sea lice”. Finally we have the answer. Apparently, the jellyfish release “grenades” made up of mucus,



containing balls of stinging cells. These structures are now called “cassiosomes”. The jellyfish release these structures into the water when they are feeding, or when agitated.

These grenades actually have outer shells that are made up of the same cells as the jellyfish, and some even have cilia, which allow them to move! Surprisingly, at the hearts of the grenades were tiny growths of symbiotic algae. Within the cassiosomes, scientists found three different kinds of toxins, that were surprisingly efficient at killing small marine animals, including shrimp. <https://dgit.in/stingingwater>



## Aliens looking at us

Scientists on Earth detect exoplanets by measuring slight dips in the brightness of their host stars as the planets pass in front of them. Now, amongst the identified planets that could detect Earth using the same method, scientists are looking for techno-signatures! <https://dgit.in/alientp>



## Hot Jupiter near death

The exoplanet with the shortest orbital period is a Jupiter-sized gas giant racing around its star, where an entire year lasts only 18 Earth hours. The planet is 1000 light years away, and scientists expect the star to swallow it in about a decade's time. <https://dgit.in/18hrplnt>



## The inside of a neutron star

Neutron stars are the still burning and extremely dense remains of massive stars that have exploded. Researchers from the University of Riken have developed a better model of the interior, which holds up even after two neutron stars collide and merge into one. <https://dgit.in/nsmodel>